Understanding SQL Tables: Permanent and Temporary

📊 Types of Tables in Databases
- Two main types: permanent tables and temporary tables.
- Permanent tables live indefinitely until explicitly dropped.
- Temporary tables store intermediate results during a session and are automatically dropped once the session ends.
- Temporary tables are sometimes called shortcut temp tables.

Permanent Tables
- Created with CREATE and INSERT or CTAS (Create Table As Select).
- Persist across sessions and system restarts.
- Used for main data storage such as customers, orders, employees, etc.

Temporary Tables
- Created during an active session, stored in temporary storage (tempdb).
- Automatically removed when the session ends (e.g., client disconnect or shutdown).
- Useful for intermediate data manipulation without impacting the source tables.

🛠️ Creating and Using Temporary Tables
- Syntax uses a hash (#) prefix before the table name to designate a temporary table.
- Created by SELECT ... INTO #temp_table FROM source_table.
- Accessible only within the session in which they were created.
- Can be queried, updated, deleted, and inserted just like permanent tables during the session.

Example Workflow
- Copy the original table data into a temporary table.
- Modify or filter data in the temporary table (e.g., delete delivered orders).
- Perform analysis or transformations on the temporary data.
- Optionally save the results back into a permanent table for future use.

🧹 Lifecycle and Storage of Temporary Tables
- Stored physically in tempdb, a system database.
- Metadata stored in the system catalog.
- Data available for the duration of the session.
- Automatic cleanup by the database engine after session ends — no manual dropping needed.

🔄 Use Cases and Best Practices
- Ideal for intermediate steps in ETL (Extract, Transform, Load) processes.
- Enable data transformations without affecting source data integrity.
- Useful for small projects or quick data manipulation "playgrounds."
- For important intermediate results, use permanent tables, views, or CTEs instead.
- Temporary tables reduce manual cleanup overhead.

📚 Summary of SQL Table Concepts
- Tables store data in rows and columns, similar to spreadsheets.
- Permanent tables persist until dropped; temporary tables are session-bound.
- Two creation methods:
  - CREATE + INSERT (two-step)
  - CTAS (one-step, based on query results)
- Views differ as they store query logic, not data.
- Temporary tables provide automatic cleanup and ease for transient data needs.
- Use CTAS for performance optimization and snapshotting data for bug analysis.

Mind Map Outline

📊 Types of SQL Tables
- Permanent Tables
  - Long-lived in database
  - Created via CREATE/INSERT or CTAS
  - Used for core data like customers, orders
- Temporary Tables
  - Session-limited lifespan
  - Stored in tempdb
  - Automatically dropped after session ends

🛠️ Creating Temporary Tables
- Syntax: SELECT ... INTO #temp_table FROM source
- Hash (#) prefix indicates temporary table
- Created during active session
- Accessible like normal tables during session

Example Usage
- Copy data from permanent table
- Modify/delete rows in temp table
- Query temp table for analysis
- Optionally save results permanently

🧹 Temporary Table Lifecycle
- Stored in tempdb system database
- Metadata in system catalog
- Exists only during session
- Automatically cleaned up on disconnect

🔄 Use Cases for Temporary Tables
- Intermediate results in ETL pipelines

- Data transformation playgrounds
- Avoid altering source tables
- Not recommended for important intermediate data (use permanent tables or views)
- Simplifies cleanup and session management

📚 Summary and Best Practices
- Tables store structured data in rows and columns
- Two table creation methods: CREATE/INSERT and CTAS
- Views vs tables: views store logic, tables store data
- Temporary tables for transient data needs
- Use CTAS for snapshotting and performance
- Prefer CTEs or views for quick intermediate results where applicable